

Python Foundations - Homework Exercises

Based on the topics covered during the lecture, please complete the following exercises. Write the Python code required to solve each problem, in your NOTEBOOKS.

Exercise 1: Basic Input and Type Casting

Write a program that asks the user for their current age in years. Calculate and print the approximate year they were born (assuming the current year is 2026). Ensure you properly cast the input string into an integer.

Exercise 2: Algorithmic Operators (Modulo)

Write a program using the modulo operator (%) to determine if a given number of days is a perfect number of weeks. Prompt the user for a number of days, and print the remaining leftover days after dividing into full 7-day weeks.

Exercise 3: Isolating Variables (Dynamic Financial Calculator)

A service company charges a flat base fare plus a variable rate per hour. Given the final total cost paid by a client, write a program that reverse-engineers the math to output exactly how many hours they were billed for. Cast the currency inputs to floats.

Exercise 4: Formatting with f-strings

Prompt the user for their first name, last name, and favorite programming language. Use a single f-string to print the following message: 'Hello, [First Name] [Last Name]! I see your favorite language is [Language].'

Exercise 5: F-Strings and Math

Ask the user for a product name, its price, and the quantity they wish to buy. Calculate the total cost. Use an f-string to print a receipt message that includes the product name and the total cost formatted to 2 decimal places (e.g., \$15.50).